

Fully automated ordering and labeling of ICA components

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Abstract—Functional magnetic resonance imaging (fMRI) is a brain imaging technique which provides detailed insights into brain function and its disruption in various brain disorders. fMRI data can be analyzed using data-driven or region-of-interest based methods. The data-driven analysis of brain activity maps involves several steps, the first of which is identifying whether the maps capture what might be interpreted as intrinsic connectivity networks (ICNs) or artifacts. This is followed by linking the ICNs to known anatomical and/or functional parcellations. Optionally, as in the study of functional network connectivity (FNC), rearranging the connectivity graph is also necessary for systematic interpretation. Here we present a toolbox that automates all these processes under minimal or no supervision with high accuracy. We provide a pretrained cross-validated elastic-net regularized general linear model for the noisecloud toolbox to separate the ICNs from artifacts. We include several well-known anatomical and functional parcellations from which researchers can choose to label the activity maps. Finally, we integrate a method for maximizing the within-domain modularity to generate a more systematically structured FNC matrix. We improve upon and integrate existing techniques and new methods to design this toolbox which can take care of all the above needs. Specifically, we show that our pretrained model achieves 89% accuracy and 100% precision at classifying ICNs from artifacts in a validation dataset. Researchers are generating brain imaging data and analyzing brain activity at an ever-increasing rate. The Autolabeller toolbox can help automate such analyses for faster and reproducible research.

Index Terms—fMRI, brain, anatomical, functional, parcellation, connectivity

I. INTRODUCTION

Worldwide brain imaging studies are generating diverse and large sample data at a rapidly increasing rate. Broad, collaborative efforts such as the Brain Research through Advancing Innovative Neurotechnologies (BRAIN) initiative are funding projects that produce massive amounts of data. But the potential benefits of such efforts cannot be fully realized without tools that can easily share, pool, and analyze the data. It is also critical to develop and automate tools to process and analyze the data to advance neuroscience.

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Functional magnetic resonance imaging (fMRI) is a widely-used technique for imaging human brain activity. fMRI can localize the variation in blood-oxygen-level-dependent (BOLD) response due to tasks or external stimuli, or at rest. However, the signal measured by fMRI is mixed with non-neural and physiological sources of variability [1]. De-noising methods for fMRI data can be broadly divided into two categories—those that use external physiological recordings or estimates of events which might produce artifacts (e.g. motion) and those that use data-driven techniques. But the precision of the former category depends on the availability and quality of the physiological measurements that are also not related to other common artifacts such as thermal noise and head movement. Data-driven approaches, on the other hand, make minimal to no assumptions about the relationship between the sources of noise and the resulting change in magnetic resonance imaging (MRI) signal and can be effective in mitigating multiple types of artifacts at once [1].

Independent component analysis (ICA) is a widely used approach for de-noising fMRI and provides a powerful technique for decomposing fMRI data into maximally independent spatial components [2]. In a typical ICA decomposition, some components clearly indicate BOLD signals, whereas the others indicate artifactual processes. fMRI analyses can be meaningfully interpreted only after identifying, subtracting, or regressing out the artifacts. Manually classifying the independent components (ICs) is time-consuming, hard to reproduce, and requires expertise [3]. Therefore, several approaches have been developed for automatic classification using spatio-temporal features of the ICs [4], [5].

Correlating the focus of activation in functional imaging studies with structural/anatomical information is important for interpreting results. There are various anatomical atlases, methods, and tools available for this purpose. One of the most well-known atlases is the Automated Anatomical Labeling (AAL) atlas, which used a volumetric labeling technique to manually separate region of interests (ROIs) for a single subject brain in common stereotaxic space [6]. Another important aspect of fMRI studies is to examine the functional architecture of the human brain. Any of the several popular

functional atlases can be used to determine the functional associations of IC activation maps. Yeo et al. used a clustering approach to identify functionally coupled networks across the cerebral cortex and released two well-known parcellations (7 networks and 17 networks) [7]. Gordon et al. generated a parcellation of putative cortical areas using resting-state functional correlations [8]. Inter-atlas variability can be concerning and Doucet et al. attempted to address this by providing the Consensual Atlas of REsting-state Networks (CAREN) atlas based on some of the most reliable atlases [9].

Spatial ICA analysis of fMRI data is often succeeded by the study of the temporal relationship between the ICNs or functional network connectivity (FNC) [10]. The FNC matrix essentially constitutes the adjacency matrix of a complete, weighted, undirected graph that can be further analyzed using graph algorithms [11]. Given a set of ICNs with different functional labels, it is desirable to reorder the graph adjacency matrix to group ICNs with the same label together. Also, it is desirable to maximize edges with higher weights closer to the main diagonal of the adjacency matrix. This allows us to easily distinguish patterns from the FNC matrix and make meaningful observations. The Brain Connectivity Toolbox (BCT) includes an algorithm for reordering FNC matrices for this purpose [12].

We describe in the paper the results of our goal to design a toolbox which can automate the above tasks. As discussed already, there are existing tools that can perform each of these tasks individually. Compared to those, the Autolabeller toolbox provides three distinct advantages. First, it incorporates all of the above tasks and provides seamless integration with the widely used group ICA for fMRI toolbox (GIFT) [13]. Second, it includes an efficient pretrained model for separating spatial ICNs from artifacts in the fMRI data. The model parameters are supplied with the toolbox and new testing data can be classified without having to re-train the model. And finally, it can also function as a standalone software (i.e., without the GIFT toolbox); the user only needs to provide fMRI volumes in an acceptable format (e.g. Neuroimaging Informatics Technology Initiative (NIFTI)) with an optional mask.

II. METHODS

A. Data & Preprocessing

We demonstrate the functionality of the Autolabeller toolbox using three resting-state fMRI datasets that are briefly described next. We obtained the first/primary dataset from the Function Biomedical Informatics Research Network (FBIRN) phase-III study [14]. This dataset includes resting-state fMRI data collected from 186 controls and 176 age and gender-matched schizophrenia patients (SZs). We retained data from 314 subjects (163 controls, mean age 36.9 years, 46 females and 151 SZs, mean age 37.8 years, 37 females) for further analysis after preprocessing and quality control. Further details about the data acquisition, preprocessing and quality control can be found in prior studies [14], [15].

We used resting-state fMRI data from the Centers of Biomedical Research Excellence (COBRE) study as the second/validation dataset under the experiment [16]. This study contained resting-state fMRI data from 100 controls and 87 SZs. The same quality control and preprocessing procedure were applied to both primary (FBIRN) and validation (COBRE) datasets. A total of 164 subjects (82 controls, mean age 37.7 years, 19 females and 82 SZs, mean age 38 years, 17 females) were retained for further analysis.

The third and fourth validation datasets were obtained from multiple fMRI studies on large-sample controls in the Human Connectome Project (HCP) (<http://www.humanconnectomeproject.org/>) and the Genomic Superstruct Project (GSP) [17]. Preprocessed data from these projects were downloaded and then resliced to $3mm \times 3mm \times 3mm$ spatial resolution using Statistical Parametric Mapping (SPM12).

B. Deriving Spatial Maps Using Group ICA

The simplest way to use the Autolabeller is to execute it with the group ICA session information file from the GIFT software as the input [13]. Fig. 1 outlines the basic functionality of the Autolabeller. The GIFT toolbox was used to identify spatial ICs of brain activity from four different datasets (FBIRN and COBRE, HCP and GSP) separately in prior studies [15], [18], [19]. In spatial group ICA, preprocessed fMRI data of each subject are first reduced from time $\times$ voxel dimension to $PC1 \times$ voxel dimension using principal component analysis (PCA), where $PC1$ is the number of principal components (PCs). Then all subjects' data are concatenated along the time or $PC1$ dimension, and the data are reduced again to $PC2 \times$ voxel dimension using PCA. Subsequently, spatial ICs are identified from the reduced data using an ICA algorithm, such as infomax [20]. It is also necessary to perform the group ICA post-processing step in the GIFT toolbox by generating a report, as it also computes the FNC matrix. The aggregate ICN spatial maps (SMs) and corresponding ground truth labels of all four studies under consideration are included with the Autolabeller code [21].

C. Identifying ICNs

We integrated the noisecloud toolbox into the Autolabeller to distinguish ICNs from artifacts [22]. The noisecloud toolbox implements an elastic-net regularized general linear model (GLM) with 10-fold cross-validation. It has to be trained with high-quality data for reasonable performance which we generated as follows. We obtained the labeled (ICN/noise) group-level IC SMs using the primary (FBIRN) dataset from prior work [15]. Next, we obtained the subject-level IC SMs from its group ICA result. The number of group-level ICs was 100, out of which 47 were ICNs, and the number of subjects was 314. Hence we had a dataset of 31400 ICs at our disposal, out of which 14758 were labeled ICNs and 16642 were artefactual components or noises.

It was computationally expensive and time-consuming to train the model using all of the data, and the performance

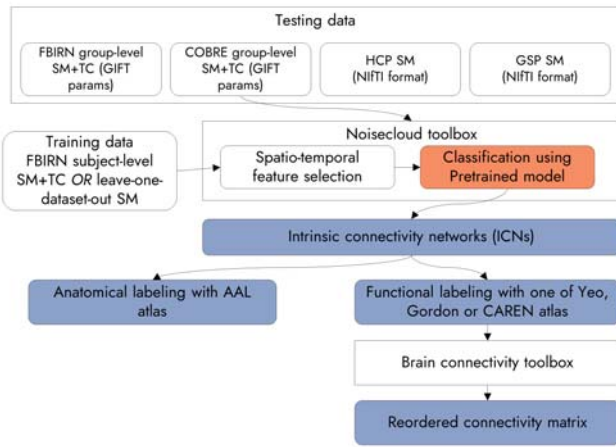


Fig. 1: Flowchart of our analysis using the Autolabeller toolbox. SM stands for spatial map and TC for time course. Three sets of inputs are used, two of them are group ICA results from the GIFT toolbox, and one in Nifti format. The integration of noisecloud and BCT toolboxes is shown [12], [22]. A pretrained model for the noisecloud toolbox, a contribution of this work, is indicated in orange. The output from the Autolabeller toolbox is indicated in purple.

gain was marginal. Therefore we randomly sampled 3000 volumes out of 31400, which we used as training data for the noisecloud toolbox. Approximately 47% of the training data were ICNs and the rest were noises. The time courses (TCs) corresponding to the subject-level IC SMs were also fed into the model for training. If the input is a Nifti volume instead of a GIFT parameter file, the TC features are not used and a model trained solely on SM features is used. The noisecloud toolbox extracts approximately 246 spatio-temporal features from the training data to train the model and uses corresponding features from the testing data to predict the ICN/noise label of the ICs. We used the mean group-level SMs, 100 each, of both the primary (FBIRN) and validation (COBRE) datasets as the testing data. The latest version of the noisecloud toolbox comes bundled with the GIFT toolbox, but can also be used in a standalone manner.

In addition to the above model trained with subject-level IC features, we trained another model using group-level IC features (aggregate SMs from the GIFT toolbox) from four datasets (FBIRN, COBRE, HCP, and GSP). In this case, the model was trained in a leave-one-dataset-out manner, i.e., a model was trained using group-level IC characteristics from three datasets (with 10-fold cross-validation), and then tested using the group-level IC features of the remaining dataset. The aggregate SMs have arbitrary scale which were thresholded by a value of 5.

D. Anatomical Labeling of SMs

We determined the anatomical label of a region of activation by correlating its SM with the masks of the AAL atlas regions

[6], [23]. At first, the testing data (FBIRN and COBRE mean group ICA components) were resampled to the same space as the AAL atlas using the SPM12 toolbox [24]. Each region in the AAL atlas was converted to a binary mask. The pairwise correspondence between the AAL masks and volumes in the testing data was established using the Pearson correlation function in the Matlab software. Autolabeller provides two more options for determining the correspondence, i.e., Matthews correlation and cluster labeling. The top three (a number which the user can control) AAL region masks with the highest degree of correspondence with each volume are retained as the result.

E. Functional Labeling of SMs

We determined the functional label of a region of activation in the same manner as in anatomical labeling. The functional parcellations currently available in the Autolabeller include the Yeo 2011 functional parcellations (17 networks version) in conjunction with the Buckner functional cerebellar parcellation [7], [25], Gordon 2016 [8], and CAREN [9].

F. Reordering of the FNC Matrix

The Autolabeller integrates functionality from BCT for reordering the FNC matrix [12]. Specifically, we used the `reorder_mod` function from this toolbox to reorder the FNC matrix. This function can be applied to both binary and weighted as well as on directed and undirected graphs. It utilizes the graph community structure to reorder nodes so that the edges with higher weights are closer to the main diagonal of the graph adjacency matrix. Once the optimal order is determined, the output of the previous steps (artifact detection, anatomical, and functional labeling) are all automatically rearranged in the same order.

III. RESULTS

A. Identifying ICNs

The Autolabeller incorporates a pretrained cross-validated elastic-net regularized GLM for separating ICNs from artifacts. Tab. I demonstrates the performance of the pretrained models applied to different datasets using two different training samples. The accuracy, precision and recall values for recognizing ICNs are all in percentages. The model predicted ICN labels with 85% accuracy in the primary (FBIRN) dataset and 89% accuracy in a validation (COBRE) dataset using both the SMs and TCs as features. A 100% precision in the validation dataset indicates that none of the true ICNs were wrongly labeled as artifacts. ICN labels were predicted with 89% accuracy in both the primary and the validation (COBRE) datasets using only the SMs as features. The asterisk (*) indicates that the testing data was biased, i.e., FBIRN mean or aggregate ICA components were tested using a model trained with FBIRN subject-level IC features. In the validation datasets where only the SM features were used, i.e., HCP and GSP, the accuracy was still reasonable (86% and 77% respectively).

TABLE I: Performance in detecting ICN vs. noise using a pretrained model

Testing data	Training data	Training N	Train. acc.	Test. acc.	Precision	Recall
FBIRN mean SM+TC	FBIRN sub. SM+TC	3000	87.19	85	80.85	86.36
COBRE mean SM+TC		3000	87.19	89	100	76.59
FBIRN agg. SM	Leave-one-dataset	300	74	89	80.85	95
COBRE agg. SM	-out SM	300	71.67	89	94.44	79.07
HCP agg. SM		300	74.33	86	78.43	93.02
GSP agg. SM		300	85	77	66.67	85

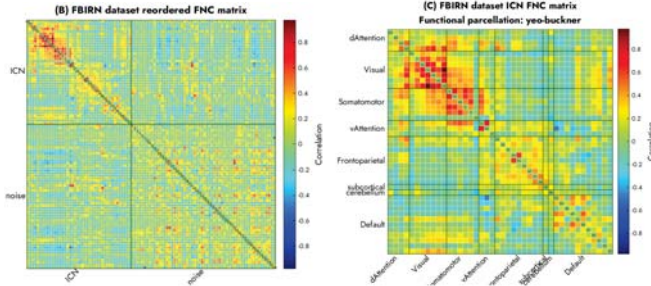


Fig. 2: Unordered vs. automatically labeled and reordered FNC matrices. The Autolabeller separates resting-state ICNs from noise and groups the ICNs into functional domains with high modularity.

B. Anatomical & Functional Labeling

The supplementary tables list the anatomical and functional labels of the ICN SMs from the FBIRN and COBRE datasets. For each fMRI volume, the output indicates whether it is an ICN or not, and lists the top three highest correlated ROIs in the chosen anatomical/functional parcellation. The tables list the labels determined by the Autolabeller as well as those from prior works (ground truths). For the FBIRN dataset, 32 out of 47 (68.08%) of both the anatomical and functional labels agree with the labeling from the prior work [15]. For the COBRE dataset, 25 out of 36 (69.44%) of both the anatomical and functional labels agree with the labeling from the prior work [18].

C. Reordering of the FNC Matrix

Fig. 2 displays the FNC matrices for the unordered components (left), separate ICN and noise components (middle), and ordered ICNs (right) in the FBIRN (top) and COBRE (bottom) datasets. The ICNs on the right are grouped into functional domains based on the Yeo 2011 atlas and Buckner cerebellar parcellation. The FNC matrices are ordered by the Autolabeller to increase the modularity of the community structures within each functional domain. As such, the reordered matrices are much easier to visualize and interpret.

IV. DISCUSSION

Separating artifacts from ICNs, labeling the foci of functional activation, and modularizing the functional connectivity are some of the tasks still many researchers perform manually. Here we design a toolbox that can enable automation of these tasks for improving brain imaging studies. It provides

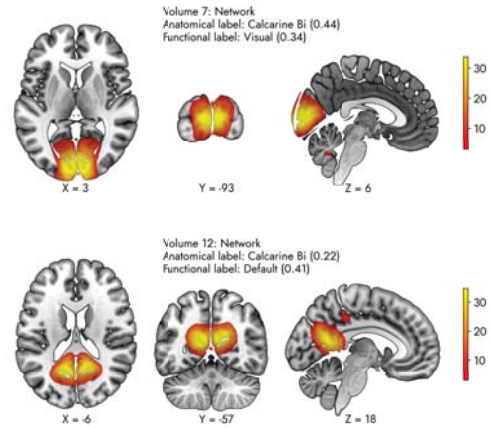


Fig. 3: Examples of Autolabeller prediction in FBIRN dataset

the user with relevant anatomical and functional correlates of functional activation, which facilitate further analysis of neuroimaging results. The toolbox can be seamlessly integrated with group ICA analysis in the GIFT toolbox, but can also be used in a standalone fashion with fMRI data in an acceptable format (e.g. Nifti) and can incorporate an optional brain mask.

Next, we discuss some of the subtle differences between the Autolabeller output and the ground truths that are based on prior studies. Fig. 3 displays the contrast between the anatomical/functional labels of some of the SMs. Volume 7 (left) is labeled as a bilateral calcarine component, as it has a correlation value of 0.44 with the left Calcarine ROI and 0.39 with the right Calcarine ROI. The Autolabeller output (provided with the code) lists right Lingual gyrus the 3rd highest correlated (0.13) ROI. However, in the prior work, it is labeled as right Cuneus [15]. On the other hand, volume 12 (right) is labeled as bilateral Calcarine by both the Autolabeller and the prior work, but in terms of function, it is associated with default mode network (DMN) and visual networks respectively. According to the Autolabeller output, this volume is highly correlated with multiple DMN ROI of the Yeo 2011 atlas (top 3 correlation values of 0.41, 0.4, and 0.25).

We retained the noisecloud toolbox in the Autolabeller but generated a model ourselves for using in conjunction with it. The noisecloud toolbox, available as part of the GIFT toolbox, already includes an algorithm that can extract 246

different spatial and temporal features from the input data. This is one of the largest feature sets in the literature for this particular problem, which makes for a robust algorithm capable of generalizing well and detecting multiple types of noise components. Our model was trained using a subset of the subject-level SMs and TC features from the primary dataset (FBIRN). During prediction, it achieved higher accuracy in the validation dataset (COBRE) compared to the primary dataset, which proves that the model generalizes very well. Although we provide a pretrained model, the user still has the option to provide their training data, which allows for more flexibility and better sensitivity especially when a large volume of training data is available.

The Autolabeller integrates well-known atlases such as AAL, Yeo 2011, Gordon 2016, and CAREN for analyzing the anatomical and functional correlates of brain activity. By using these labeling, researchers can publish reproducible results instantly comparable to other studies using any of these same atlases. The results show that compared to prior work using the same primary and validation datasets, the accuracy of anatomical and functional labeling is rather low (below 70%). This is however expected because prior works used subjective criteria for labeling instead of quantitatively determining association with a known atlas [26]. Another limitation of the Autolabeller is a limited number of atlas choices. In particular, not every one of these may cover the entire cortical, subcortical, and cerebellar regions of the brain. The Yeo 2011 functional atlas choice in the autolabeller in fact integrates both the Yeo 2011 cortical parcellation and Buckner 2011 cerebellum parcellation. However, Gordon 2016 or CAREN atlases do not include cerebellum parcellation. We will add support for more atlases, both anatomical and functional in the future to alleviate these problems.

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